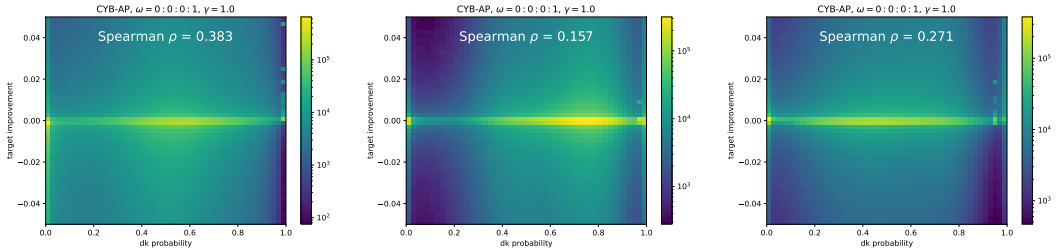

Figure 10: Density plots indicating that <DON'T KNOW> probability is positively correlated with improvement in accuracy, for steps 0, 1, and 2.

token can also be variable depending on the context. In Figure 12, which contains many digits, it's apparent that the model does not slow down when it is predicting a string of digits.

Table 3 shows pause-count statistics for selected tokens for a model trained with CYB-AP. The Table presents tokens which tend to be followed by few ('low median') or many ('high median') pauses. The 'low variance' row indicates tokens which tend to have the same number of pauses regardless of their context; the 'high variance' row indicates tokens for which the model's pausing behavior is context dependent.

7 Discussion

In this article, we introduced a family of losses that enable foundation models to set their own pace of generation. We examined three variants of the *Catch Your Breath* (CYB) loss family and converged on one particular setting that results in models that dynamically request additional computation steps when it will benefit output quality. The CYB losses reflect a synergy between two important research threads in machine learning, one focused on developing adaptive-computation techniques for LLMs and another on leveraging non-content control tokens in the input stream (here, <PAUSE> tokens). The synergy produces greater benefit from <PAUSE> tokens relative to earlier approaches that were based on the standard cross-entropy loss; and the synergy provides a more practical means of adapting computation time than dynamic-depth approaches.

The CYB loss obtains an explicit measure of uncertainty via the <DON'T KNOW> output. This uncertainty is epistemic, meaning that the model has not yet acquired sufficient information. Ordinarily, epistemic uncertainty refers to limitations of *training*, but in the case of CYB, it refers to limitations of *computation*: the model is uncertain because it has not yet performed sufficient compute operations.

We proposed two distinct recipes for training and inference with CYB losses, but we focused on the recipe in which a fixed number of pauses are inserted after each token during both training and inference. The alternative, in which pauses are inserted only when the model requests them, requires running an LLM autoregressively. It would thus make most sense to introduce variable pauses at the

Table 3: **Statistics of pause counts for selected tokens following CYB training**. Examples are presented of tokens which tend to be followed by few (*low median*) or many (*high median*) pauses. The *low variance* row indicates tokens which tend to have the same number of pauses regardless of their context; the *high variance* row indicates tokens for which the model's pausing behavior is context dependent.

	Low Median	**High Median**
Low Variance	isn, wasn, didn, doesn, don, according, able, co, However, plenty, etc, Inc, &, non, addition	devices, players, students, projects, applications, challenges, environment, systems, stories, games, groups, events, patients, families, teams
High Variance	g, e, to, entry, S, won, co, New, Y, V, F, Z, World, of, Air	materials, images, @, site, pictures, photos, sites, code, College, wood, page, file, professionals, is, homes

Figure 11: Model trained with CYB loss produces variable pause durations on tokens. Color indicates the expected number of processing steps to predict the corresponding token. When the model produces high probability <DON'T KNOW> outputs, it receives additional processing steps.

post-training (RL) alignment stage where autoregressive updating is performed. Beyond these two recipes, other possibilities exist, such as developing strategies to predict which tokens are likely to most benefit and inserting pauses selectively.

One direction of future work is to better understand how model's utilize the additional pause steps. From one perspective, the model's input during these pause steps is nearly identical to that of the initial step. To compute different features during each delay step, the model must index on the embedding that is associated with each pause index. One might consider facilitating the model's use of pauses by conditioning some of the attention-head weights on the pause index, providing the model with a more direct means of pause-conditioned compute modulation.

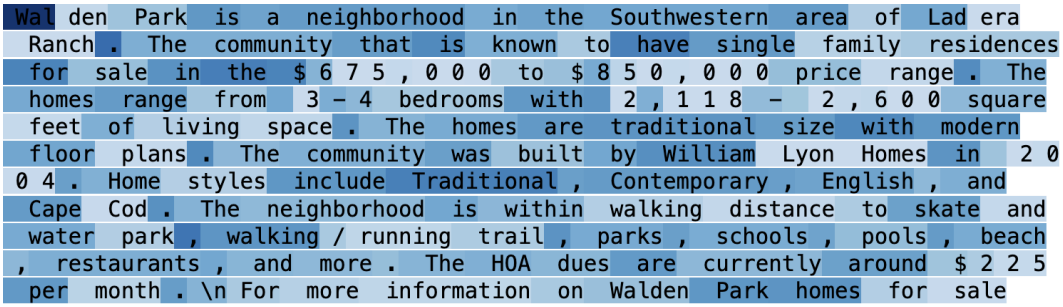

Figure 12: Model trained with CYB loss produces variable pause durations on tokens.

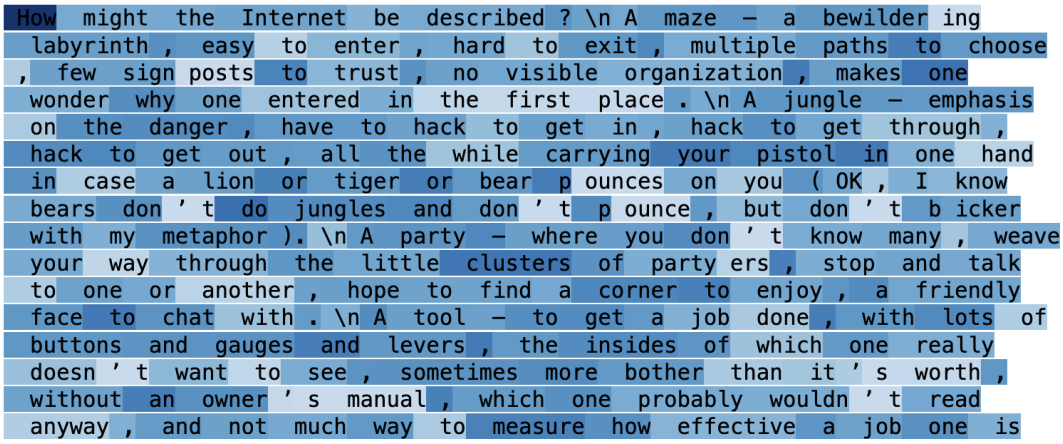

Figure 13: Model trained with CYB loss produces variable pause durations on tokens.